NOW RECRUITING • GEORGE W. WOODRUFF SCHOOL OF MECHANICAL ENGINEERING

Fully Funded PhD Positions in Experimental Fluid Mechanics

Where fundamental physics meets engineering impact

ABOUT THE GABBARD LAB

The Gabbard Lab is an experimental fluid mechanics and soft matter group in the George W. Woodruff School of Mechanical Engineering at Georgia Tech. Our mission is to uncover the fundamental mechanisms of flowing and soft matter to improve engineering systems. We design, build, and run novel tabletop experiments, paired with physics-focused analysis. Our findings inform industrial heat and mass transfer, guide new environmental remediation techniques, explain natural phenomena, and advance liquid manipulation in capillary-dominated systems.

AREAS OF INTEREST



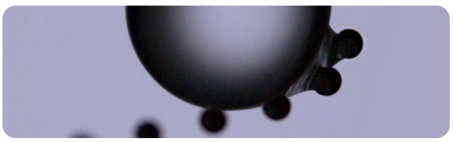
Generation and manipulation of complex droplets

How do polymers, adjuvants, and surfactants influence the mechanisms governing spray atomization, droplet-on-demand technologies, and droplet-substrate interactions?



Dynamic wetting transitions

How do features of real-world surfaces, such as elasticity, temperature, and microstructure, influence dynamic wetting transitions in droplet and thin film flows?



Particle-interface interactions

How can we improve particle scavenging, oil remediation techniques, and particle transport in systems with flowing or flexible interfaces?

WHO SHOULD APPLY

- A B.S. or M.S. in mechanical, aerospace, or chemical engineering, physics, applied math, or a related field
- Curiosity about fluid mechanics, soft matter, and the physics of everyday phenomena
- Enthusiasm for hands-on work: building setups, machining, 3D printing, or electronics
- Some programming for data and image analysis (e.g., Python or MATLAB), or eagerness to learn
- Clear communication, teamwork, and persistence

We are always eager to add new skillsets beyond these to our group!

WHY GEORGIA TECH

Georgia Tech is a research powerhouse, ranked #2 in the nation for federally sponsored research with $1.43 billion in research expenditures (FY2025), where students work alongside world-class faculty at the cutting edge of science and engineering. Its campus sits in Midtown Atlanta, a vibrant, diverse city and well-connected global hub.

BY THE NUMBERS

#1 Public Graduate Mechanical Engineering Program (#3 overall)

#2 Public Graduate Engineering Program (#4 overall)

Rankings: U.S. News & World Report, Best Graduate Schools 2026. Research expenditures: NSF HERD Survey (2024 data) and Georgia Tech (FY2025).

HOW TO APPLY

1. **Explore our research.** Look through our current and past research and publications to make sure the Gabbard Lab aligns with your interests.
2. **Email the PI.** Send your CV, a few sentences about the research you would like to do, and the email addresses of two professional references to chase.gabbard@me.gatech.edu.
3. **Apply to Georgia Tech.** Submit a PhD application to the George W. Woodruff School of Mechanical Engineering by December 15, 2026 and mention the Gabbard Lab in your application. Apply here ↗

DATES TO REMEMBER

REACH OUT TO PI
Before December 15th

APPLY TO GEORGIA TECH
December 15, 2026

ANTICIPATED START DATE
Fall 2027

CONTACT

Chase T. Gabbard Principal Investigator
chase.gabbard@me.gatech.edu

To learn more, visit our website and Google Scholar.